Unit

38 Gas volume calculations

Pre-study

A pre-study worksheet on this unit is available for download from *Jing Kung Chemistry Website*.

Unit Key Concepts

- Calculations involving mass, volume and number of moles of a gas
- Calculations involving chemical reactions with gaseous reactants and products

38.1 The relationship between gas volume and moles: Avogadro's Law

In 1811, the Italian Amedeo Avogadro put forward the following **hypothesis**:

✓ Equal volumes of gases at the same temperature and pressure contain equal numbers of molecules.

The volume of a gas varies with temperature and pressure. Thus, the comparison is made under the same temperature and pressure.

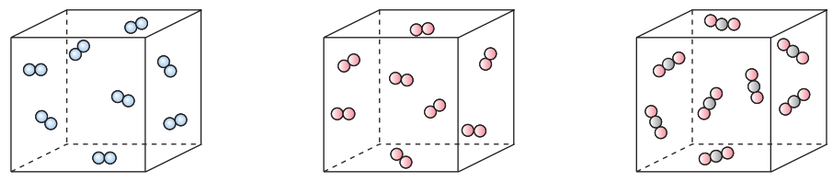


Fig. 38.1 Equal volumes of hydrogen, oxygen and carbon dioxide at the same temperature and pressure contain equal numbers of molecules

Fig. 38.2 shows a stamp quoting the Avogadro's original hypothesis.



Fig. 38.2 In 1956, Avogadro's original hypothesis was commemorated in an Italian stamp

hypothesis 假說

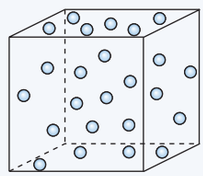
Do you know

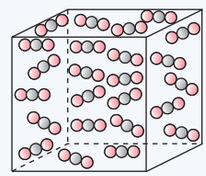
Density of a gas

Density is the mass of an object divided by its volume.

$$\text{Density of a gas} = \frac{\text{mass}}{\text{volume}}$$

The following diagram shows two containers of equal volumes containing helium and carbon dioxide gases at the same temperature and pressure. As equal volumes of gases contain equal numbers of molecules, the mass of gas in the second container is higher because the relative molecular mass of carbon dioxide is higher. Thus, the density of carbon dioxide is higher than that of helium.





	helium (He)	carbon dioxide (CO_2)
Relative molecular mass:	4.0	44.0
Density (g cm^{-3}):	0.00018	0.001977

Suppose a helium gas sample contains 1% of carbon dioxide gas, the mass and density of this impure helium sample will be higher than those of pure helium.

This idea, now known as **Avogadro's Law**, is important because it means that we can compare the number of molecules in different gases by simply comparing their volumes.

The number of molecules in one mole of any gas is always 6.02×10^{23} and equal numbers of moles of gases contain equal numbers of molecules. Avogadro's Law can also be stated as:

✓ Equal volumes of gases at the same temperature and pressure contain equal numbers of moles of gases.

38.2 Molar volume of a gas

One of the major implications of Avogadro's Law is that one mole of any gas must occupy the same volume under the same conditions. This is the molar volume of the gas.

Avogadro's Law 亞佛加德羅定律

Do you know?

From hypothesis to law

Most scientific advances start off with a hypothesis — a tentative explanation for an observation. It can even be just an educated guess.

Once the hypothesis has been tested repeatedly for many years and is found to hold true by several different scientists, it may become accepted as a law.

Avogadro's hypothesis was at first neglected by scientists, partly because it contradicted the views of the respected scientist John Dalton. Avogadro was the first person to have the idea that atoms grouped together into molecules. Dalton thought that matter was just made of atoms. It was not until after Avogadro's death in 1854 that his brilliant deduction was accepted by the scientific community. Avogadro's hypothesis became a law.

✓ The volume occupied by one mole of a gas is called the **molar volume**.

CE 2006 Paper 2 Q12

We can determine the molar volume of hydrogen gas using the experimental set-up shown in Fig. 38.3.

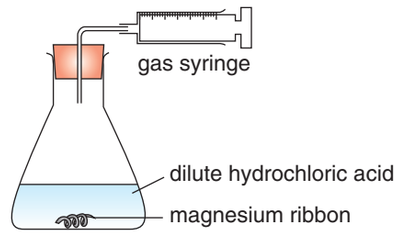


Fig. 38.3 Experimental set-up for determining the molar volume of hydrogen gas

1 Place excess dilute hydrochloric acid in a conical flask.

2 Weigh a piece of clean magnesium ribbon. Record its mass.

3 Put the magnesium ribbon into the dilute hydrochloric acid.

4 Record the volume of hydrogen gas released.

The following data are obtained in a typical experiment:

Mass of magnesium ribbon reacted = 0.100 g

Volume of hydrogen gas released = 98.8 cm^3

Let V dm^3 be the molar volume of hydrogen gas.

Magnesium reacts with dilute hydrochloric acid according to the following equation:

$$Mg(s) + 2HCl(aq) \longrightarrow MgCl_2(aq) + H_2(g)$$

$$\text{Number of moles of magnesium reacted} = \frac{0.100\ g}{24.3\ g\ mol^{-1}}$$

According to the equation, 1 mole of Mg reacts with 2 moles of HCl to give 1 mole of H_2.

$$\text{i.e. number of moles of hydrogen gas released} = \frac{0.100\ g}{24.3\ g\ mol^{-1}}$$

$$= \frac{\frac{98.8}{1\,000}\ dm^3}{V\ dm^3}$$

$$V = 24.0\ dm^3$$

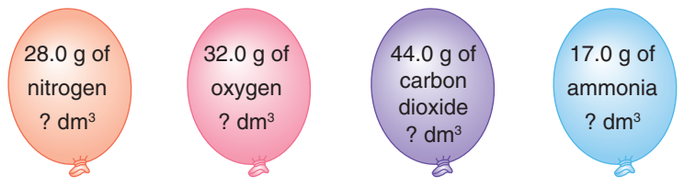
molar volume 摩爾體積

N1 Refer to page T24.

At 25 °C and 1 atmospheric pressure, the volume occupied by 1 mole of hydrogen gas is 24.0 dm^3. This is the molar volume of hydrogen gas. N1

Practice 38.1

Work out the molar volume of the following gases (at room temperature and pressure).



Gas	Mass of 1 mole of gas	Mass of 1 dm^3 of gas	Molar volume
N_2	28.0 g	1.166 g	24.0 dm^3
O_2	32.0 g	1.333 g	24.0 dm^3
CO_2	44.0 g	1.833 g	24.0 dm^3
NH_3	17.0 g	0.708 g	24.0 dm^3

38.1

Determining the molar volume of carbon dioxide.

The molar volume of any gas at 25 °C and 1 atmosphere (at room temperature and pressure) is 24.0 $dm^3\ mol^{-1}$ (or 24 000 $cm^3\ mol^{-1}$) (Fig. 38.4).

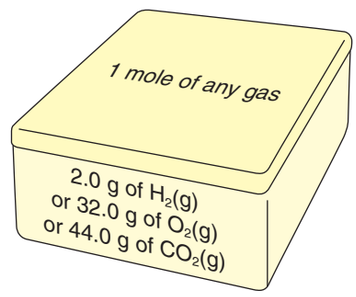


Fig. 38.4 A mole of any gas at room temperature and pressure occupies 24.0 dm^3 (about the volume of a large biscuit tin)

We can calculate the number of moles of a gas using the following formula:

✓ Under the same temperature and pressure,

$$\text{Number of moles of gas} = \frac{\text{volume of gas}}{\text{molar volume of gas}}$$

Fig. 38.5 summarizes the steps in calculations involving the mass, number of moles and volume of a gas.

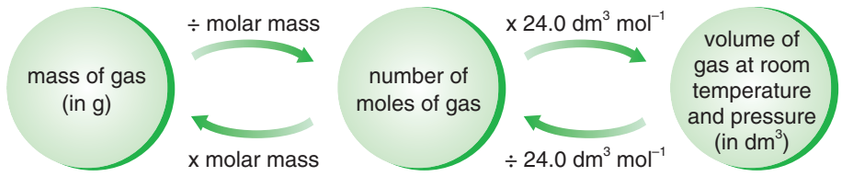


Fig. 38.5 The relationship involving the mass, number of moles and volume of a gas in calculations

Example 38.1

Q What is the number of moles of methane in 1.44 dm^3 of the gas measured at room temperature and pressure?

(Molar volume of gas at room temperature and pressure = 24.0 $dm^3\ mol^{-1}$)

A Number of moles of methane

$$= \frac{\text{volume of methane (at room temperature and pressure)}}{\text{molar volume of gas (at room temperature and pressure)}}$$

$$= \frac{1.44\ dm^3}{24.0\ dm^3\ mol^{-1}}$$

$$= 0.0600\ mol$$

$\therefore$ 1.44 dm^3 of methane contain 0.0600 mole of the gas.

Example 38.2

Q What is the number of molecules in 1 200 cm^3 of nitrogen measured at room temperature and pressure?

(Molar volume of gas at room temperature and pressure = 24 000 cm^3 mol^{-1}; Avogadro constant = 6.02 x 10^{23} mol^{-1})

A Number of moles of nitrogen

$$= \frac{\text{volume of } N_2 \text{ (at room temperature and pressure)}}{\text{molar volume of gas (at room temperature and pressure)}}$$

$$= \frac{1\ 200\ cm^3}{24\ 000\ cm^3\ mol^{-1}}$$

$$= 0.0500\ mol$$

Number of nitrogen molecules
= 0.0500 mol x 6.02 x 10^{23} mol^{-1}
= 3.01 x 10^{22}

∴ 1 200 cm^3 of nitrogen contain 3.01 x 10^{22} molecules.

Example 38.3

Q What is the volume, measured at room temperature and pressure, occupied by 13.2 g of carbon dioxide?

(Relative atomic masses: C = 12.0, O = 16.0; molar volume of gas at room temperature and pressure = 24.0 dm^3 mol^{-1})

A Number of moles of CO_2 $= \frac{13.2\ g}{44.0\ g\ mol^{-1}}$

$= 0.300\ mol$

Volume of CO_2 (at room temperature and pressure)
= number of moles of CO_2 x molar volume of gas (at room temperature and pressure)
= 0.300 mol x 24.0 dm^3 mol^{-1}
= 7.20 dm^3

∴ 13.2 g of carbon dioxide occupy 7.20 dm^3.

Example 38.4

Q An oxide of metal M reacts completely with carbon to give 25.9 g of metal M and 3.00 dm^3 of carbon dioxide measured at room temperature and pressure. What is the empirical formula of the oxide?

(Relative atomic masses: O = 16.0, M = 207.2; molar volume of gas at room temperature and pressure = 24.0 $dm^3\ mol^{-1}$)

A Number of moles of CO_2 = $\frac{\text{volume of } CO_2 \text{ (at room temperature and pressure)}}{\text{molar volume of gas (at room temperature and pressure)}}$

$= \frac{3.00\ dm^3}{24.0\ dm^3\ mol^{-1}}$

$= 0.125$ mol

i.e. number of moles of oxygen atoms in the oxide = 2 x 0.125 mol
= 0.250 mol

Number of moles of M in the oxide = $\frac{25.9\ g}{207.2\ g\ mol^{-1}}$
= 0.125 mol

Number of moles of M : number of moles of O = 0.125 : 0.250
= 1 : 2

∴ the empirical formula of the oxide is MO_2.

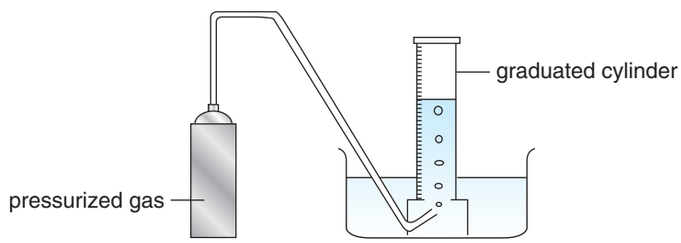
Practice 38.2

Refer to page T25.

1 What is the volume, measured at room temperature and pressure, occupied by 6.80 g of ammonia?
9.60 dm^3
(Relative atomic masses: H = 1.0, N = 14.0; molar volume of gas at room temperature and pressure = 24.0 $dm^3\ mol^{-1}$)

2 What is the number of moles of chlorine in 672 cm^3 of the gas measured at room temperature and pressure?
0.0280 mole
(Molar volume of gas at room temperature and pressure = 24 000 $cm^3\ mol^{-1}$)

3 The diagram below shows a set-up used to find the molar mass of a gas.



The gas container was weighed and then a certain volume of the gas was dispersed into the graduated cylinder. The gas container was then reweighed.

Continued on next page

The following results were obtained in the experiment:

Initial mass of gas container = 78.98 g

Volume of gas dispersed into graduated cylinder = 600 cm^3

Final mass of gas container = 77.53 g

Calculate the molar mass of the gas.

58.0 g mol^{-1}

(Molar volume of gas at room temperature and pressure = 24.0 dm^3 mol^{-1})

38.3 Calculations from chemical equations

Using a balanced chemical equation and certain information, we can calculate the amounts of the reactants consumed or products formed in a reaction. The steps are as follows (Fig. 38.6):

1 Write down a balanced chemical equation for the reaction.

2 Write down the amounts of substances given in the question underneath their formulae.

3 Convert each given amount to number of moles.

4 Using the balanced chemical equation, work out the number of moles of the substance required in the question.

5 Convert the number of moles of the substance required to mass (g), volume (dm^3) or concentration (mol dm^{-3}).

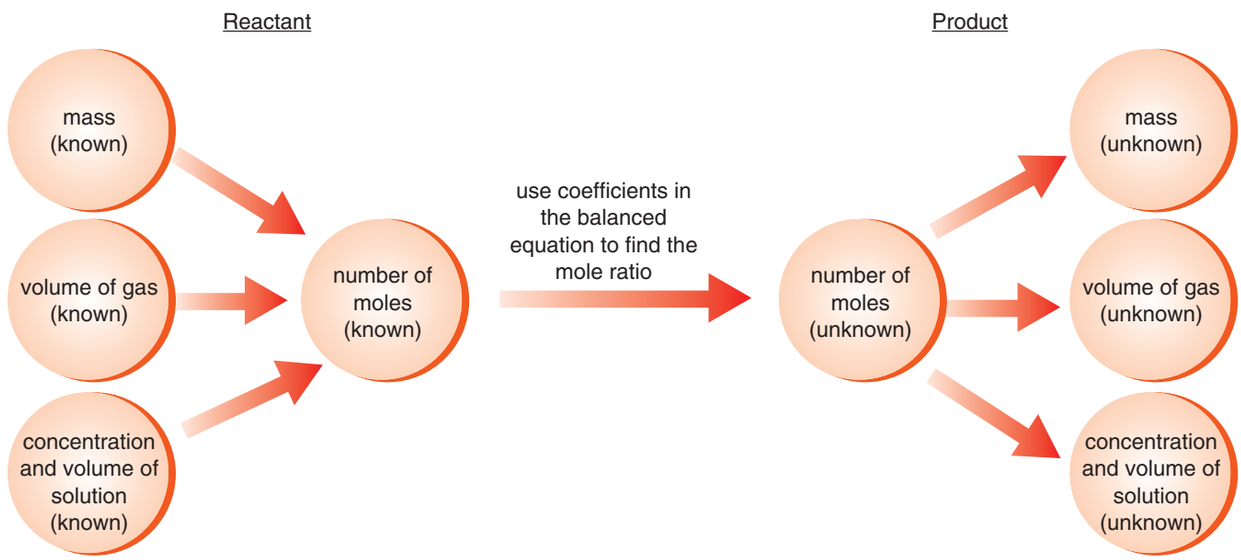


Fig. 38.6 Steps in calculations involving chemical equations

Example 38.5

Q The decomposition of sodium azide, NaN_3, takes place when an airbag inflates. An airbag requires a large volume of a gas to be produced in a few milliseconds. The gas is produced by the rapid decomposition of the sodium azide.

$$2NaN_3(s) \longrightarrow 2Na(l) + 3N_2(g)$$

What is the volume of gas produced, measured at room temperature and pressure, when 78.0 g of NaN_3 decompose?

(Relative atomic masses: N = 14.0, Na = 23.0; molar volume of gas at room temperature and pressure = 24.0 $dm^3\ mol^{-1}$)

DSE 2013 Paper 1B Q11(b)

A Number of moles of NaN_3 decomposed $= \dfrac{78.0\ g}{65.0\ g\ mol^{-1}}$

$= 1.20\ mol$

According to the equation, 2 moles of NaN_3 decompose to give 3 moles of N_2.

i.e. number of moles of N_2 produced $= 1.20 \times \dfrac{3}{2}\ mol$

$= 1.80\ mol$

Volume of N_2 produced (at room temperature and pressure)
= number of moles of N_2 x molar volume of gas (at room temperature and pressure)
= 1.80 mol x 24.0 $dm^3\ mol^{-1}$
= 43.2 dm^3

$\therefore$ 43.2 dm^3 of nitrogen gas are produced.

Example 38.6

Q 0.400 g of an impure sample of magnesium ribbons reacts with excess dilute sulphuric acid to give 312 cm^3 of hydrogen, measured at room temperature and pressure.

a) What is the percentage by mass of magnesium in the sample?

b) State ONE assumption in the calculation.

(Relative atomic mass: Mg = 24.3; molar volume of gas at room temperature and pressure = 24.0 $dm^3\ mol^{-1}$)

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A a) Magnesium and dilute sulphuric acid react according to the following equation:

$$Mg(s) + H_2SO_4(aq) \longrightarrow MgSO_4(aq) + H_2(g)$$

$$\text{Number of moles of } H_2 = \frac{\text{volume of } H_2 \text{ (at room temperature and pressure)}}{\text{molar volume of gas (at room temperature and pressure)}}$$

$$= \frac{\frac{312}{1\,000}\ dm^3}{24.0\ dm^3\ mol^{-1}}$$

$$= 0.0130\ mol$$

According to the equation, 1 mole of Mg gives 1 mole of H_2 with dilute sulphuric acid.

i.e. number of moles of Mg reacted = 0.0130 mol

$$\text{Mass of Mg in sample} = 0.0130\ mol \times 24.3\ g\ mol^{-1}$$
$$= 0.316\ g$$

$$\text{Percentage by mass of Mg in sample} = \frac{0.316\ g}{0.400\ g} \times 100\%$$
$$= 79.0\%$$

∴ the percentage by mass of magnesium in the sample is 79.0%.

b) The sample does not contain other metals that give hydrogen with dilute sulphuric acid.

Example 38.7

Q 12.0 g of calcium carbonate are added to 100.0 cm^3 of 1.00 mol dm^{-3} hydrochloric acid. What is the volume of carbon dioxide liberated at room temperature and pressure?

(Relative atomic masses: C = 12.0, O = 16.0, Ca = 40.1; molar volume of gas at room temperature and pressure = 24.0 dm^3 mol^{-1})

CE 2010 Paper 2 Q37
DSE 2012 Paper 1A Q25

A

$$\underset{12.0\ g}{CaCO_3(s)} + \underset{\substack{1.00\ mol\ dm^{-3}\\ 100.0\ cm^3}}{2HCl(aq)} \longrightarrow CaCl_2(aq) + \underset{?\ cm^3}{CO_2(g)} + H_2O(l)$$

$$\text{Number of moles of } CaCO_3 \text{ present} = \frac{12.0\ g}{100.1\ g\ mol^{-1}}$$
$$= 0.120\ mol$$

$$\text{Number of moles of HCl present} = 1.00\ mol\ dm^{-3} \times \frac{100.0}{1\,000}\ dm^3$$
$$= 0.100\ mol$$

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According to the equation, 1 mole of $CaCO_3$ reacts with 2 moles of HCl to produce 1 mole of CO_2. $CaCO_3$ is in excess in this case. The amount of HCl limits the amount of CO_2 liberated.

$$\text{Number of moles of } CO_2 \text{ liberated} = \frac{0.100}{2} \text{ mol}$$
$$= 0.0500 \text{ mol}$$

Volume of CO_2 liberated (at room temperature and pressure)
= number of moles of CO_2 x molar volume of gas (at room temperature and pressure)
= 0.0500 mol x 24.0 dm^3 mol^{-1}
= 1.20 dm^3

$\therefore$ 1.20 dm^3 of carbon dioxide are liberated.

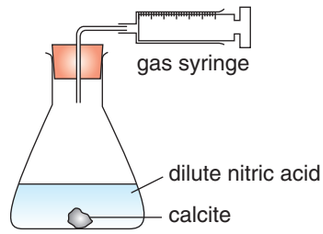
Practice 38.3

1 Upon heating, silver oxide decomposes according to the following equation:

$2Ag_2O(s) \longrightarrow 4Ag(s) + O_2(g)$

What is the volume of $O_2(g)$ formed at room temperature and pressure if 11.6 g of silver oxide undergo complete decomposition?
0.600 dm^3; refer to page T26.
(Relative atomic masses: O = 16.0, Ag = 107.9; molar volume of gas at room temperature and pressure = 24.0 dm^3 mol^{-1})

2 Calcite is a mineral which contains mainly calcium carbonate. The set-up shown below was used to determine the percentage by mass of calcium carbonate in a sample of calcite.



0.380 g of calcite was allowed to react with excess nitric acid. 84.0 cm^3 of gas were collected in the syringe.

a) Suggest how one could know that the reaction between calcite and nitric acid had completed.
No more gas was produced.
b) Calculate the percentage by mass of calcium carbonate in the sample of calcite.
92.1%; refer to page T26.
c) State ONE assumption in the calculation.
The sample of calcite does not contain other carbonates that give carbon dioxide with nitric acid.
(Relative atomic masses: C = 12.0, O = 16.0, Ca = 40.1; molar volume of gas at room temperature and pressure = 24.0 dm^3 mol^{-1})

38.4 Gas volume – gas volume calculations from chemical equations

Example 38.8

Q Butane burns in oxygen completely to produce carbon dioxide and water according to the following equation:

$$2C_4H_{10}(g) + 13O_2(g) \longrightarrow 8CO_2(g) + 10H_2O(l)$$

What volume of oxygen is required for the complete combustion of 480 cm^3 of butane (measured at room temperature and pressure)?

(Molar volume of gas at room temperature and pressure = 24.0 $dm^3\ mol^{-1}$ = 24 000 $cm^3\ mol^{-1}$)

A Method 1

$$2C_4H_{10}(g) + 13O_2(g) \longrightarrow 8CO_2(g) + 10H_2O(l)$$
$$480\ cm^3 \qquad ?\ cm^3$$

Number of moles of C_4H_{10}

$$= \frac{\text{volume of } C_4H_{10} \text{ (at room temperature and pressure)}}{\text{molar volume of gas (at room temperature and pressure)}}$$

$$= \frac{480\ cm^3}{24\ 000\ cm^3\ mol^{-1}}$$

$$= 0.0200\ mol$$

According to the equation, 2 moles of C_4H_{10} require 13 moles of O_2 for complete combustion.

i.e. number of moles of O_2 required $= 0.0200 \times \frac{13}{2}$ mol

$$= 0.130\ mol$$

Volume of O_2 required (at room temperature and pressure)
= number of moles of O_2 x molar volume of gas (at room temperature and pressure)
= 0.130 mol x 24.0 $dm^3\ mol^{-1}$
= 3.12 dm^3

$\therefore$ 3.12 dm^3 of oxygen are required for the complete combustion of the butane.

Continued on next page

Method 2

$$\underset{480\ cm^3}{2C_4H_{10}(g)} + \underset{?\ cm^3}{13O_2(g)} \longrightarrow 8CO_2(g) + 10H_2O(l)$$

According to the equation, 2 moles of C_4H_{10} require 13 moles of O_2 for complete combustion.

$\therefore$ 2 x 24 000 cm^3 of C_4H_{10} require 13 x 24 000 cm^3 of O_2 for complete combustion.

$$\underset{\substack{2 \times 24\,000\ cm^3 \\ 480\ cm^3}}{2C_4H_{10}(g)} + \underset{\substack{13 \times 24\,000\ cm^3 \\ ?\ cm^3}}{13O_2(g)} \longrightarrow 8CO_2(g) + 10H_2O(l)$$

Volume of O_2 required (at room temperature and pressure) $= 480\ cm^3 \times \dfrac{13 \times 24\,000\ cm^3}{2 \times 24\,000\ cm^3}$

$= 3\,120\ cm^3$

$= 3.12\ dm^3$

$\therefore$ 3.12 dm^3 of oxygen are required for the complete combustion of the butane.

Example 38.9

Q 60.0 cm^3 of carbon monoxide were allowed to burn completely in 60.0 cm^3 of oxygen. Assuming that all volumes were measured at the same temperature and pressure, what was the final gaseous volume at the end of the combustion?

A Carbon monoxide burns in oxygen according to the following equation:

$$\underset{60.0\ cm^3}{2CO(g)} + \underset{60.0\ cm^3}{O_2(g)} \longrightarrow 2CO_2(g)$$

According to the equation, 2 moles of CO react with 1 mole of O_2 to produce 2 moles of CO_2.

$\therefore$ 60.0 cm^3 of CO react with 30.0 cm^3 of O_2 to produce 60.0 cm^3 of CO_2.

Volume of O_2 left after combustion $= (60.0 - 30.0)\ cm^3$

$= 30.0\ cm^3$

Final gaseous volume = volume of O_2 left + volume of CO_2 produced

$= (30.0 + 60.0)\ cm^3$

$= 90.0\ cm^3$

$\therefore$ the final gaseous volume at the end of the combustion is 90.0 cm^3.

Practice 38.4

1 A gaseous mixture consists of methane and ethane in a mole ratio of 1 : 1. Its volume is 200 cm^3 at room temperature and pressure.

a) Write chemical equations for the complete combustion of methane and ethane.

b) Calculate the volume of oxygen required, measured at room temperature and pressure, for the complete combustion of the mixture.

(Molar volume of gas at room temperature and pressure = 24.0 $dm^3\ mol^{-1}$)

2 Gases X and Y react to give a gaseous product Z. The reaction can be represented by the equation:

$X(g) + 3Y(g) \longrightarrow 2Z(g)$

In an experiment, 40 cm^3 of X and 90 cm^3 of Y were mixed and allowed to react in a closed vessel. What was the volume of the resultant gaseous mixture?

(All volumes were measured at the same temperature and pressure.)

Practice 38.4

1 a) $CH_4(g) + 2O_2(g) \longrightarrow CO_2(g) + 2H_2O(l)$ ..... (1)

$C_2H_6(g) + \frac{7}{2} O_2(g) \longrightarrow 2CO_2(g) + 3H_2O(l)$ ..... (2)

b) According to equation (1), 1 mole of CH_4 requires 2 moles of O_2 for complete combustion.

i.e. 24 000 cm^3 of CH_4 require 2 x 24 000 cm^3 of O_2 for complete combustion.

$CH_4(g)$ + $2O_2(g)$ $\longrightarrow$ $CO_2(g)$ + $2H_2O(l)$

24 000 cm^3 2 x 24 000 cm^3

100 cm^3 ? cm^3

Volume of O_2 required = 100 cm^3 x $\frac{2 \times 24\,000\ cm^3}{24\,000\ cm^3}$

= 200 cm^3

According to equation (2), 1 mole of C_2H_6 requires $\frac{7}{2}$ moles of O_2 for complete combustion.

i.e. 24 000 cm^3 of C_2H_6 require $\frac{7}{2}$ x 24 000 cm^3 of O_2 for complete combustion.

$C_2H_6(g)$ + $\frac{7}{2} O_2(g)$ $\longrightarrow$ $2CO_2(g)$ + $3H_2O(l)$

24 000 cm^3 $\frac{7}{2}$ x 24 000 cm^3

100 cm^3 ? cm^3

Volume of O_2 required = 100 cm^3 x $\frac{\frac{7}{2} \times 24\,000\ cm^3}{24\,000\ cm^3}$

= 350 cm^3

Total volume of O_2 required = (200 + 350) cm^3

= 550 cm^3

2 $X(g) + 3Y(g) \longrightarrow 2Z(g)$

40 cm^3 90 cm^3

According to the equation, 1 mole of X reacts with 3 moles of Y to produce 2 moles of Z.

∴ 30 cm^3 of X react with 90 cm^3 of Y to produce 60 cm^3 of Z, i.e. 10 cm^3 of X would remain.

∴ volume of the resultant gaseous mixture = (10 + 60) cm^3

= 70 cm^3

Have you mastered?

Key terms

hypothesis 假說 70 Avogadro's Law 亞佛加德羅定律 71 molar volume 摩爾體積 72

Checklist

After studying this unit, you should be able to

- ☐ deduce the molar volume of gases at room temperature and pressure as 24.0 dm^3 using given data;
- ☐ perform calculations involving the mass, volume and number of moles of a gas;
- ☐ perform calculations involving chemical reactions with gaseous reactants and products.

(Put a '✔' in the box if you have acquired the knowledge concerned.)

Summary

1 Avogadro's Law states that equal volumes of gases at the same temperature and pressure contain equal numbers of molecules.

2 Equal volumes of gases at the same temperature and pressure contain equal numbers of moles of gases.

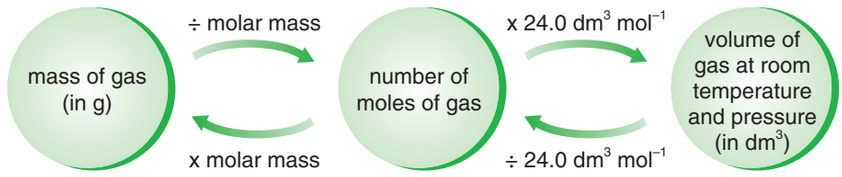
3 The volume occupied by one mole of a gas is called the molar volume.

4 At room temperature and pressure (25 °C and 1 atmospheric pressure), the molar volume of any gas is 24.0 $dm^3\ mol^{-1}$ (or 24 000 $cm^3\ mol^{-1}$).

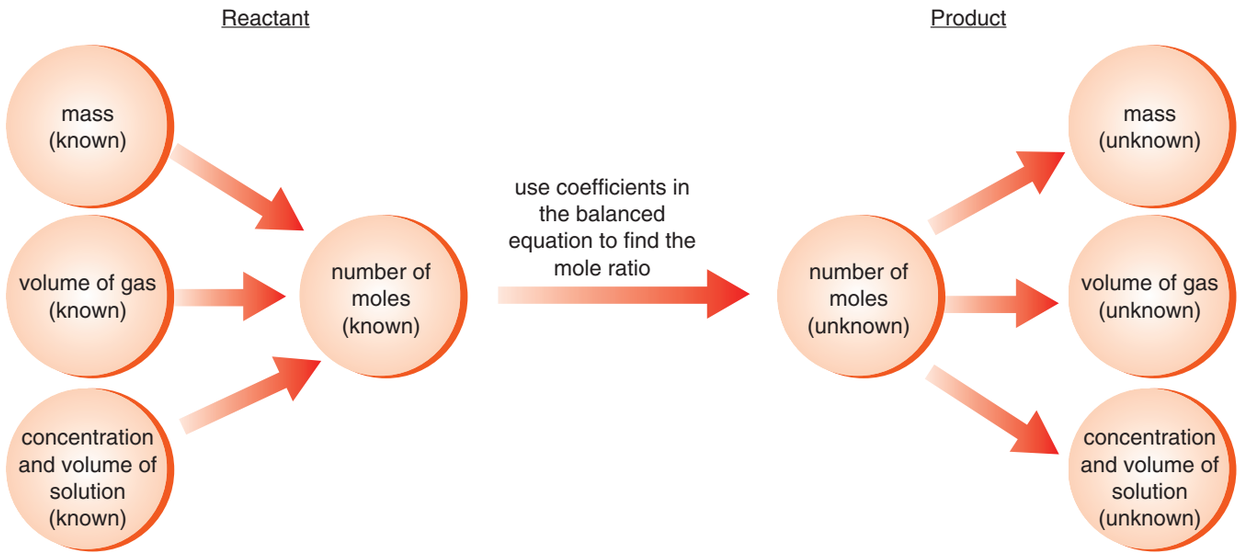
5 Under the same temperature and pressure,

$$\text{Number of moles of gas} = \frac{\text{volume of gas}}{\text{molar volume of gas}}$$

6 The relationship involving the mass, number of moles and volume of a gas is as follows:



7 Below are the steps used in calculations involving chemical reactions with gaseous reactants and products:



Have you mastered?

Unit Exercise

Note: The symbol ! indicates the level of difficulty of a question.

Part I Knowledge and understanding

1 Fill in the blanks.

a) Number of moles of gas = $\dfrac{\text{volume of gas}}{\text{molar volume of gas}}$

b) mass of gas (in gram) ⇄ (÷ molar mass / x molar mass) number of moles of gas ⇄ (x 24.0 dm^3 mol^{-1} / ÷ 24.0 dm^3 mol^{-1}) volume of gas at room temperature and pressure (in dm^3)

2 Complete the following table.

Formula of gas	Mass of 1 mole of gas	Mass of gas in sample	Number of moles of gas in sample	Volume of gas sample at room temperature and pressure
CH_4	16.0 g	4.00 g	$\frac{4.00\text{ g}}{16.0\text{ g mol}^{-1}}$ = 0.250 mol	0.250 mol x 24.0 dm^3 mol^{-1} = 6.00 dm^3
NO_2	46.0 g	0.125 mol x 46.0 g mol^{-1} = 5.75 g	$\frac{3.00\text{ dm}^3}{24.0\text{ dm}^3\text{ mol}^{-1}}$ = 0.125 mol	3.00 dm^3
NH_3	17.0 g	1.25 mol x 17.0 g mol^{-1} = 21.3 g	1.25 mol	1.25 mol x 24.0 dm^3 mol^{-1} = 30.0 dm^3
Cl_2	71.0 g	3.55 g	$\frac{3.55\text{ g}}{71.0\text{ g mol}^{-1}}$ = 0.0500 mol	0.0500 mol x 24.0 dm^3 mol^{-1} = 1.20 dm^3

(Relative atomic masses: H = 1.0, C = 12.0, N = 14.0, O = 16.0, Cl = 35.5; molar volume of gas at room temperature and pressure = 24.0 dm^3 mol^{-1})

Part II Multiple choice questions

3 Which of the following gas samples occupies the greatest volume at the same temperature and pressure?

Refer to page T27 for explanation.

(Relative atomic masses: H = 1.0, C = 12.0, O = 16.0, F = 19.0, Ne = 20.0)

A 1 gram of ethane
B 1 gram of oxygen
C 1 gram of fluorine
(D) 1 gram of neon

(Edexcel Advanced Subsidiary GCE, Unit 1, Jun. 2009, 17)

4 At room temperature and pressure, if 1 dm^3 of hydrogen chloride gas contains x atoms, then 1 dm^3 of methane gas contains

A $\frac{2}{5}x$ atoms.
(B) $\frac{5}{2}x$ atoms.
C $2x$ atoms.
D $5x$ atoms.

5 The equation for the complete combustion of octane is

$$2C_8H_{18} + 25O_2 \longrightarrow 16CO_2 + 18H_2O$$

The volume of 1 mole of any gas (measured at room temperature and pressure) is 24.0 dm^3. Hence the volume of oxygen (measured at room temperature and pressure) required for the complete combustion of 10.0 moles of octane is

A 240 dm^3.
B 300 dm^3.
(C) 3 000 dm^3.
D 6 000 dm^3.

(Edexcel Advanced Subsidiary GCE, Unit 1, Jan. 2011, 4(b))

6 In an experiment, excess calcium granules are added to 100.0 cm^3 of 2.0 M hydrochloric acid. What is the theoretical volume of hydrogen gas liberated at room temperature and pressure?

(Molar volume of gas at room temperature and pressure = 24.0 $dm^3\ mol^{-1}$)

A 0.6 dm^3
B 1.2 dm^3
(C) 2.4 dm^3
D 4.8 dm^3

(HKCEE, Paper 2, 2011, 33)

7 What is the theoretical volume of carbon dioxide gas, measured at room temperature and pressure, that can be obtained by adding 100 cm^3 of 2.0 M HCl(aq) to 0.80 g of $Na_2CO_3(s)$?

(Relative atomic masses: H = 1.0, C = 12.0, O = 16.0, Na = 23.0, Cl = 35.5; molar volume of gas at room temperature and pressure = 24.0 $dm^3\ mol^{-1}$)

A 90 cm^3
(B) 180 cm^3
C 240 cm^3
D 480 cm^3

(HKCEE, Paper 2, 2010, 37)

8 Assuming that air contains 20% of oxygen by volume, how much air is required to burn completely 100 cm^3 of ethane? (All volumes are measured at the same temperature and pressure.)

A 350 cm^3
B 1 000 cm^3
(C) 1 750 cm^3
D 3 500 cm^3

(HKCEE, Paper 2, 2009, 40)

4 1 dm^3 of HCl(g) and 1 dm^3 of $CH_4(g)$ contain the same number of molecules.
One molecule of HCl(g) contains 2 atoms while one molecule of $CH_4(g)$ contains 5 atoms.
Thus, if 1 dm^3 of HCl(g) contains x atoms, then 1 dm^3 of $CH_4(g)$ contains $\frac{5}{2}x$ atoms.

5 $2C_8H_{18}(l) + 25O_2(g) \longrightarrow 16CO_2(g) + 18H_2O(l)$
10.0 mol
According to the equation, 2 moles of C_8H_{18} require 25 moles of O_2 for complete combustion.
$\therefore$ 10.0 moles of C_8H_{18} required 125 moles of O_2 for complete combustion.
Volume of O_2 required = 125 mol x 24.0 $dm^3\ mol^{-1}$
= 3 000 dm^3
$\therefore$ 3 000 dm^3 of oxygen are required for complete combustion.

9 Complete combustion of 50 cm^3 of a hydrocarbon vapour gave 350 cm^3 of carbon dioxide, both gas volumes being measured at the same temperature and pressure. The formula of the hydrocarbon could be

Refer to page T27 for explanation.

A C_8H_{18}.
B C_7H_{16}.
C C_6H_{14}.
D C_5H_{12}.

(Edexcel Advanced Subsidiary GCE, Unit 1, May 2012, 2)

10 At room temperature and pressure, 1 mole of gas A and 2 moles of gas B react completely to form 1 mole of gas C and 1 mole of gas D. If the temperature and pressure remain unchanged, which of the following will decrease after the reaction?

(1) The mass of the gaseous mixture
(2) The volume of the gaseous mixture
(3) The total number of atoms making up the gases in the gaseous mixture

A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

(HKCEE, Paper 2, 2010, 46)

Part III Structured questions

Unit Exercise

11 400.0 cm^3 of carbon dioxide gas were drawn into a gas syringe. A syringe cap was then fitted and the syringe weighed. The mass of the carbon dioxide was found to be 0.7488 g.

Refer to page T27.

a) Calculate the density (in g dm^{-3}) of the carbon dioxide gas under the experimental conditions.
1.872 g dm^{-3}
b) Calculate the volume of one mole of the gas under the experimental conditions.
23.5 dm^3 mol^{-1}
c) At the same temperature and pressure, 200.0 cm^3 of a gas G has a mass of 0.5448 g. Calculate the molar mass of G.
64.0 g mol^{-1}

(Relative atomic masses: C = 12.0, O = 16.0)

12 Ammonia can be prepared in the laboratory using the reaction between calcium hydroxide and ammonium chloride.

0.240 dm^3; refer to page T28.

$Ca(OH)_2(aq) + 2NH_4Cl(s) \longrightarrow CaCl_2(aq) + 2NH_3(g) + 2H_2O(l)$

Calculate the volume of ammonia formed at room temperature and pressure from 0.535 g of ammonium chloride.

(Relative atomic masses: H = 1.0, N = 14.0, Cl = 35.5; molar volume of gas at room temperature and pressure = 24.0 dm^3 mol^{-1})

(Edexcel GCE O Level, Paper 2, Jan. 2009, 6(b)(iii))

13 The element strontium forms a nitrate, $Sr(NO_3)_2$, which decomposes on heating as shown below.

1.50 dm^3; refer to page T28.

$2Sr(NO_3)_2(s) \longrightarrow 2SrO(s) + 4NO_2(g) + O_2(g)$

A student heats 5.29 g of $Sr(NO_3)_2$ and collects the gas at room temperature and pressure.

Calculate the volume of gas, in dm^3, obtained by the student at room temperature and pressure.

(Molar mass of $Sr(NO_3)_2$ = 211.6 g mol^{-1}; molar volume of gas at room temperature and pressure = 24.0 dm^3 mol^{-1})

(OCR Advanced Subsidiary GCE, Chem. A, F321, Jan. 2010, 5(d)(ii))

14 Sodium hydrogencarbonate decomposes upon heating to form sodium carbonate, carbon dioxide and water.

a) Write a chemical equation for the decomposition of sodium hydrogencarbonate.
$2NaHCO_3(s) \longrightarrow Na_2CO_3(s) + CO_2(g) + H_2O(l)$
b) Calculate the volume of carbon dioxide formed at room temperature and pressure if 54.6 g of sodium hydrogencarbonate undergo complete decomposition.
7.80 dm^3; refer to page T28.

(Relative atomic masses: H = 1.0, C = 12.0, O = 16.0, Na = 23.0; molar volume of gas at room temperature and pressure = 24.0 dm^3 mol^{-1})

15 Sodium azide (NaN_3) is a stable compound at room temperature but decomposes when heated to 300 °C. The equation for the decomposition is:

Refer to page T29.

$2NaN_3(s) \longrightarrow 2Na(l) + 3N_2(g)$

Sodium azide is used to produce nitrogen gas to inflate car airbags.



If a car is involved in a collision, the sodium azide decomposes. The nitrogen gas is produced very rapidly and the airbag inflates almost immediately.

a) A fully-inflated airbag has a total volume of 108 dm^3.

Calculate the amount of nitrogen, in moles, in a fully-inflated airbag.

4.50 mol

(You should assume that the volume of one mole of nitrogen inside the airbag is 24.0 dm^3.)

b) Use your answer to (a) to calculate the mass, in grams, of sodium azide required to produce 108 dm^3 of nitrogen.

195 g

(Relative atomic masses: N = 14.0, Na = 23.0)

(Edexcel IGCSE, Chemistry, Paper 2C, Jun. 2011, 7(a))

16 A metal carbonate decomposes according to the equation:

65.0; zinc; refer to page T29.

$MCO_3(s) \longrightarrow MO(s) + CO_2(g)$

When 2.50 g of MCO_3 are decomposed completely by heating, 480 cm^3 of carbon dioxide are produced at room temperature and pressure. Calculate the formula mass of MCO_3 and hence identify M.

(Relative atomic masses: C = 12.0, O = 16.0; molar volume of gas at room temperature and pressure = 24.0 $dm^3\ mol^{-1}$)

(Edexcel GCE O Level, Paper 2, Jan. 2011, 9(d))

17 An oxide of metal M reacts completely with carbon to give 38.1 g of metal M and 3.60 dm^3 of carbon dioxide measured at room temperature and pressure. What is the empirical formula of the oxide?

M_2O; refer to page T29.

(Relative atomic masses: O = 16.0, M = 63.5; molar volume of gas at room temperature and pressure = 24.0 $dm^3\ mol^{-1}$)

18 a) An impure sample of sodium hydrogencarbonate, $NaHCO_3$, of mass 0.227 g, was reacted with an excess of hydrochloric acid. The volume of carbon dioxide evolved was measured at room temperature and pressure and found to be 58.4 cm^3.

Refer to page T30.

$NaHCO_3(s) + HCl(aq) \longrightarrow NaCl(aq) + H_2O(l) + CO_2(g)$

The molar volume of any gas at the temperature and pressure of the experiment is 24.0 $dm^3\ mol^{-1}$. The molar mass of sodium hydrogencarbonate is 84.0 $g\ mol^{-1}$.

i) Calculate the number of moles of carbon dioxide given off.

0.00243 mol

ii) Calculate the mass of sodium hydrogencarbonate present in the impure sample.

0.204 g

iii) Calculate the percentage purity of the sodium hydrogencarbonate.

89.9%

b) Suggest why the carbon dioxide should NOT be collected over water in this experiment.

(Edexcel Advanced Subsidiary GCE, Unit 1, Jan. 2012, 20(a), (b)(ii))

19 What volume of oxygen is required to burn completely 200 cm^3 of propane?

1 000 cm^3; refer to page T30.

(All volumes are measured at the same temperature and pressure.)

20 Ammonia reacts with oxygen in the presence of a catalyst to give nitrogen monoxide and steam.

220 cm^3; refer to page T31.

$4NH_3(g) + 5O_2(g) \longrightarrow 4NO(g) + 6H_2O(g)$

100 cm^3 of $NH_3(g)$ and 100 cm^3 of $O_2(g)$ were mixed and allowed to react in a closed vessel. What was the volume of the resultant gaseous mixture?

(All volumes were measured at the same temperature and pressure.)

21 3.88 g of magnesium carbonate are added to 50.0 cm^3 of 2.00 mol dm^{-3} hydrochloric acid.

a) Write a chemical equation for the reaction between magnesium carbonate and hydrochloric acid.
$MgCO_3(s) + 2HCl(aq) \longrightarrow MgCl_2(aq) + CO_2(g) + H_2O(l)$

b) What is the volume of gas liberated at room temperature and pressure?
1.10 dm^3; refer to page T31.
(Relative atomic masses: C = 12.0, O = 16.0, Mg = 24.3; molar volume of gas at room temperature and pressure = 24.0 dm^3 mol^{-1})

22 a) Based on the information given below, deduce the structures of two isomeric compounds:
Refer to page T31.

- Both compounds contain carbon and hydrogen only.
- Complete combustion of 0.0150 mole of each compound gives 0.810 g of water and 1 080 cm^3 of carbon dioxide (measured at room temperature and pressure).

(Relative atomic masses: H = 1.0, O = 16.0; molar volume of gas at room temperature and pressure = 24.0 dm^3 mol^{-1})

b) Suggest a chemical test to distinguish between the two isomers in (a).

23 The first stage in the manufacture of nitric acid is the catalytic oxidation of ammonia to form nitrogen monoxide, NO.
Refer to page T32.

a) 0.150 g of nitrogen monoxide gas, NO, has a volume of 112 cm^3 at 0 °C and 1 atmosphere. Calculate the volume of one mole of the gas under these conditions.
22 400 cm^3
(Relative atomic masses: N = 14.0, O = 16.0)

b) The reaction between nitrogen monoxide and oxygen is represented by the equation:

$2NO(g) + O_2(g) \longrightarrow 2NO_2(g)$

If 200 cm^3 of nitrogen monoxide react with excess oxygen, what volume of nitrogen dioxide will be formed?
200 cm^3
(All measurements being taken at the same temperature and pressure.)

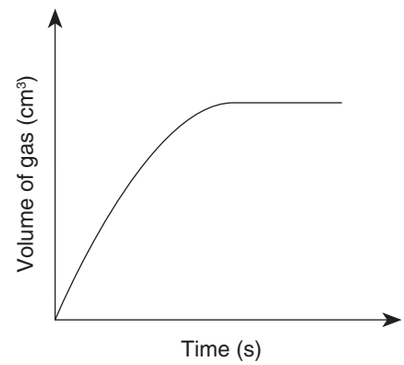
c) The Avogadro constant is represented as L. Calculate, in terms of L, the number of molecules in 480 cm^3 of nitrogen monoxide at room temperature and pressure.
0.0200 L
(Molar volume of gas at room temperature and pressure = 24 000 cm^3 mol^{-1})

(Edexcel GCE O Level, Paper 2, May 2010, 4(b), (c), (d)(ii))

24 A student investigated the rate of the reaction between zinc and dilute hydrochloric acid.
Refer to page T33.

a) Write a chemical equation for the reaction between zinc and dilute hydrochloric acid.
$Zn(s) + 2HCl(aq) \longrightarrow ZnCl_2(aq) + H_2(g)$

b) The student added excess dilute hydrochloric acid to 0.981 g of zinc. The student collected the gas and measured the volume, at regular intervals, until after the reaction was complete. She then plotted a graph of the results.



i) Use collision theory to explain the changes in the rate of the reaction as it proceeds.

ii) Calculate the volume of gas given off, measured at room temperature and pressure.
360 cm^3
(Relative atomic mass: Zn = 65.4; molar volume of gas at room temperature and pressure = 24.0 dm^3 mol^{-1})

c) The student repeated the experiment using ethanoic acid instead of hydrochloric acid. Assume the concentration of both acids and all other conditions are the same.

State and explain the effect, if any, of changing the acid in

i) the rate of the reaction;

ii) the volume of gas given off.

Unit Exercise

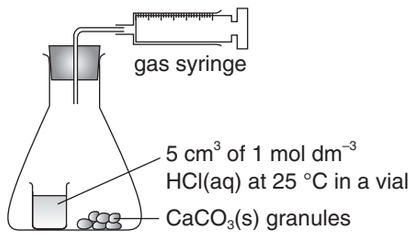
Topic Exercise

Note: The symbol 🌶 indicates the level of difficulty of a question.

Part I Multiple choice questions

1 The set-up shown below is used in an experiment to study the rate of the reaction:

$CaCO_3(s) + 2HCl(aq) \longrightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$



The conical flask is shaken to overturn the vial in order to start the reaction. The initial rate of the reaction with respect to the gas liberated is determined. The experiment is then repeated with only one of the conditions changed while the others remain unchanged.

Under which of the following situations would the initial rate be the same as that in the original experiment?

(A) Using 10 cm^3 of 1 mol dm^{-3} HCl(aq)
B Using 5 cm^3 of 2 mol dm^{-3} HCl(aq)
C Using 5 cm^3 of 1 mol dm^{-3} HCl(aq) which is preheated to 50 °C
D Using powdered $CaCO_3(s)$ of the same mass

(HKDSE, Practice paper 1A, 2012, 25)

🌶🌶🌶 2 Consider the following information on two reactions involving magnesium ribbons of the same shape:

	Reaction mixture
Reaction 1	1.5 g of Mg + 100 cm^3 of 1 M HCl
Reaction 2	1.5 g of Mg + 100 cm^3 of 1 M H_2SO_4

Which of the following statements is correct?

(Relative atomic mass: Mg = 24.3)

A The magnesium reacts completely in *Reaction 1*.
B The sulphuric acid reacts completely in *Reaction 2*.
C The initial rate of *Reaction 1* and *Reaction 2* are the same.
(D) The initial rate of *Reaction 1* is smaller than that of *Reaction 2*.

(HKCEE, Paper 2, 2008, 39)

🌶 3 The rate of chemical reactions can be increased by increasing the temperature and by increasing the concentration of reactants.

Which row correctly describes the effect on particle collisions of increasing the temperature and of increasing the concentration of reactants?

	Effect on collisions of increasing the temperature	Effect on collisions of increasing the concentration
A	same frequency but of higher energy	increased frequency but of same energy
B	increased frequency and of higher energy	increased frequency and of higher energy
C	increased frequency but of same energy	increased frequency and of higher energy
(D)	increased frequency and of higher energy	increased frequency but of same energy

(Edexcel GCSE (Higher Tier), Chemistry, Unit C2, Mar. 2009, 39)

3 Increasing the temperature increases both the frequency and energy of collisions.
Increasing the concentration increases the frequency of collision, but NOT the energy of collisions.

4 A catalyst can be added to a reaction mixture to change the rate of the reaction.

Which row shows the effect that a catalyst has on the rate of a reaction and how the mass of the catalyst after the reaction compares with the mass of catalyst added? A catalyst increases the rate of reaction, but its mass remains unchanged.

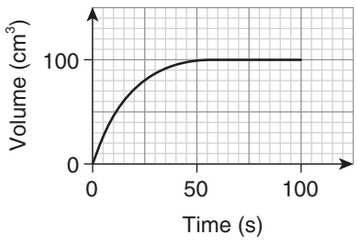
	Effect on rate of reaction	Change in mass of catalyst
(A)	increases	unchanged
B	increases	decreases
C	decreases	unchanged
D	decreases	decreases

(Edexcel GCSE (Foundation and Higher Tiers), Chemistry, Unit C2, Mar. 2011, 23)

5 In an experiment to study the rate of the following reaction, a small amount of powdered calcium carbonate was added to excess hydrochloric acid and the volume of gas liberated was recorded.

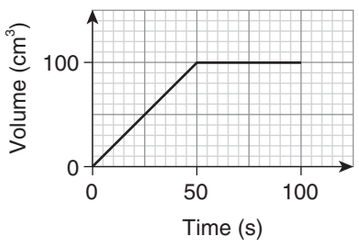
$$CaCO_3(s) + 2HCl(aq) \longrightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$$

The graph below shows the volumes of gas liberated at different times during the experiment:

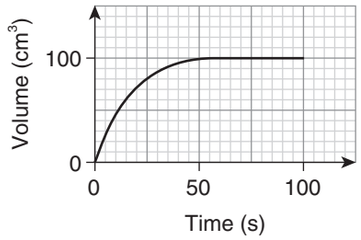


The experiment was repeated under the same conditions using the same mass of calcium carbonate granules instead of powdered calcium carbonate. Which of the following graphs would best represent the results obtained in the repeated experiment?

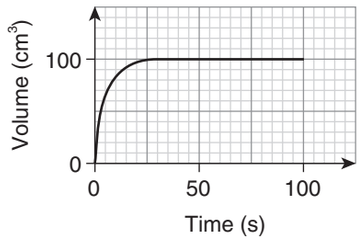
A



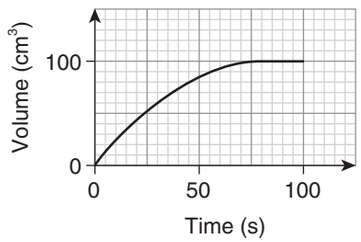
B



C



(D)



(HKDSE, Paper 1A, 2013, 25)

6 Nitroglycerin ($C_3H_5N_3O_9$) is an explosive and can explode according to the following equation:

$$4C_3H_5N_3O_9(l) \longrightarrow 12CO_2(g) + 10H_2O(g) + 6N_2(g) + O_2(g)$$

0.1 mole of nitroglycerin undergoes explosion and the products are allowed to cool to room temperature. What is the total volume of gases left behind at room temperature and pressure?

(Molar volume of gas at room temperature and pressure = 24.0 $dm^3\ mol^{-1}$)

(A) 11.4 dm^3
B 17.4 dm^3
C 45.6 dm^3
D 69.6 dm^3

(HKCEE, Paper 2, 2008, 32)

Topic Exercise

7 An oxide of metal M reacts completely with carbon to give 12.6 g of metal M and 2.38 dm^3 of carbon dioxide measured at room temperature and pressure. What is the chemical formula of the oxide?

(Relative atomic masses: M = 63.5, O = 16.0; molar volume of gas at room temperature and pressure = 24.0 $dm^3\ mol^{-1}$)

(A) MO
B MO_2
C M_2O
D M_2O_3

(HKCEE, Paper 2, 2009, 33)

8 0.40 g of an impure sample of zinc granules reacts with excess dilute sulphuric acid to give 100 cm^3 of hydrogen, measured at room temperature and pressure. Assuming that the impurities in the zinc granules do not react with sulphuric acid, what is the percentage by mass of zinc in the sample?

(Relative atomic masses: H = 1.0, Zn = 65.4; molar volume of gas at room temperature and pressure = 24.0 $dm^3\ mol^{-1}$)

A 25%
B 34%
(C) 68%
D 73%

(HKDSE, Practice paper 1A, 2012, 29)

9 What is the theoretical volume of carbon dioxide that can be obtained, at room temperature and pressure, when 1.2 g of $Na_2CO_3(s)$ reacts with 50 cm^3 of 1.0 M HNO_3?

(Relative atomic masses: H = 1.0, C = 12.0, N = 14.0, O = 16.0, Na = 23.0; molar volume of gas at room temperature and pressure = 24.0 $dm^3\ mol^{-1}$)

(A) 272 cm^3
B 544 cm^3
C 600 cm^3
D 1 200 cm^3

(HKDSE, Paper 1A, 2012, 25)

10 An important reaction which occurs in the catalytic converter of a car is

Refer to page T34 for explanation.

$2CO(g) + 2NO(g) \longrightarrow 2CO_2(g) + N_2(g)$

In this reaction, when 500 cm^3 of CO reacts with 500 cm^3 of NO at 650 °C (the operating temperature of the catalyst) and at 1 atm, the total volume of gases produced at the same temperature and pressure is

A 500 cm^3.
(B) 750 cm^3.
C 1 000 cm^3.
D impossible to calculate without knowing the molar volume of gases under these conditions.

(Edexcel Advanced Subsidiary GCE, Unit 1, May 2011, 5)

11 In an experiment to determine the initial rate of the reaction between dilute hydrochloric acid and magnesium carbonate powder, which of the following items may be measured at regular intervals as the reaction proceeds?

(1) The colour intensity of the reaction mixture
(2) The mass of the reaction mixture
(3) The volume of gas liberated

A (1) only
B (2) only
C (1) and (3) only
(D) (2) and (3) only

(HKCEE, Paper 2, 2011, 45)

12 Beaker A contains 100 cm^3 of 1 M HCl(aq), while beaker B contains 50 cm^3 of 2 M HCl(aq). Equal mass of magnesium ribbons are added to the two beakers. Both magnesium ribbons disappear after reaction. Which of the following statements is / are correct?

(1) The reactions occurring in both beakers have the same initial rate.
(2) Same volume of gas, measured at the same temperature and pressure, is given out in both beakers.
(3) Magnesium chloride solutions of the same concentration are produced in both beakers.

A (1) only
(B) (2) only
C (1) and (3) only
D (2) and (3) only

(HKCEE, Paper 2, 2009, 43)

Directions : *Each question (Questions 13–15) consists of two separate statements. Decide whether each of the two statements is true or false; if both are true, then decide whether or not the second statement is a correct explanation of the first statement. Then select one option from A to D according to the following table :*

A Both statements are true and the 2nd statement is a correct explanation of the 1st statement.
B Both statements are true but the 2nd statement is NOT a correct explanation of the 1st statement.
C The 1st statement is false but the 2nd statement is true.
D Both statements are false.

	1st statement	2nd statement	
13	All collisions between $H_2(g)$ and $I_2(g)$ molecules must lead to a reaction.	Only $H_2(g)$ and $I_2(g)$ molecules with kinetic energy equal to or greater than the activation energy can collide with each other.	D

Only $H_2(g)$ and $I_2(g)$ molecules with kinetic energy equal to or greater than the activation energy can react.

14 A catalyst can change the amount of product formed in a reaction. — A catalyst can alter the rate of a reaction. — C

A catalyst CANNOT change the amount of product formed in a reaction.

15 When equal mass of Mg and Zn granules is added separately to excess dilute H_2SO_4, a greater amount of gas will be produced by Mg than Zn. — Mg is more reactive than Zn. — B

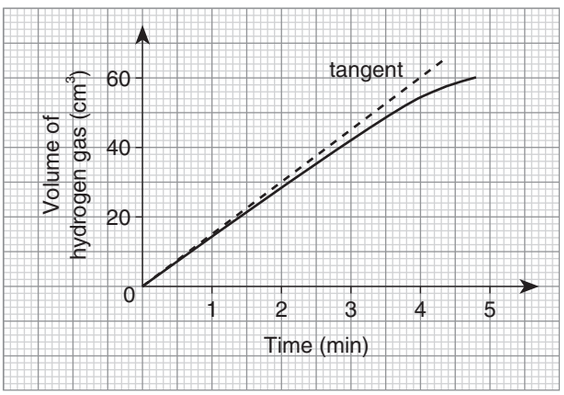
(HKCEE, Paper 2, 2008, 50)

Topic Exercise

Part II Structured questions

Answers for the questions of the public examinations in Hong Kong are not provided.

16 In an experiment, 50 cm^3 of 2.0 M HCl(aq) were added to a conical flask containing 2.0 g of zinc powder. The curve in the graph shows the volume, measured at room temperature and pressure, of the hydrogen gas liberated in the first few minutes of the experiment. The dotted line in the graph is the tangent to the curve at the start of the reaction.



a) The 'initial rate' of a reaction is defined as the instantaneous rate at the start of the reaction. With reference to the graph above, calculate the initial rate of the reaction with respect to the volume of hydrogen gas liberated.

b) Explain qualitatively the effect on the initial rate of the reaction of replacing the 2.0 M HCl(aq) with 2.0 M H_2SO_4(aq).

c) Upon completion of the reaction, all the zinc powder was used up. Calculate the theoretical volume of hydrogen gas liberated, measured at room temperature and pressure.

(Relative atomic masses: Zn = 65.4; molar volume of gas at room temperature and pressure = 24.0 $dm^3\ mol^{-1}$)

(HKDSE, Paper 1B, 2012, 11)

17 An excess of marble chips, $CaCO_3$, was added to nitric acid in a conical flask at room temperature. A plug of cotton wool was placed in the neck of the flask. The mass of the flask and its contents was weighed every minute for six minutes. The results are shown below.

Time (min)	0	1	2	3	4	5	6
Loss in mass (g)	0.00	0.56	0.87	1.00	1.06	1.08	1.08

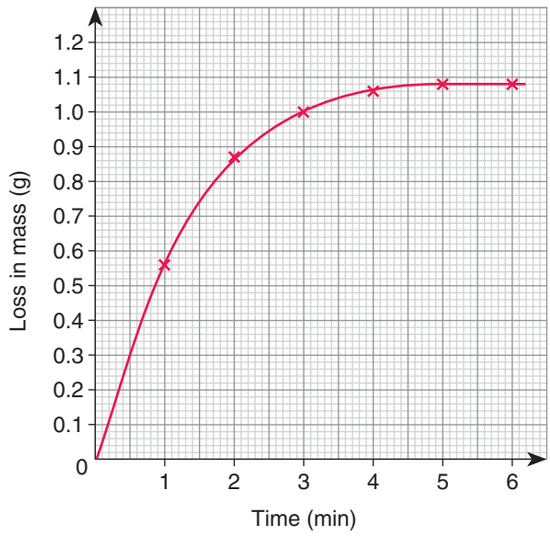
The equation for the reaction is

$$CaCO_3(s) + 2HNO_3(aq) \longrightarrow Ca(NO_3)_2(aq) + CO_2(g) + H_2O(l)$$

a) State why the mass of the flask and contents decreased.
The carbon dioxide gas escaped.

b) Why was a plug of cotton wool used?
To stop any acid from splashing out.

c) Plot the results on the grid below and draw a line of best fit through the points.



d) i) Use the graph to describe how the rate of reaction changes during the experiment.

ii) Explain in terms of collision theory why the rate changes during the experiment.

17 d) i) The rate of reaction is the highest at the start.
The rate of reaction decreases with time.

ii) As the acid particles are consumed gradually, the acid concentration decreases.
The chance of collision decreases and the number of effective collisions in a unit volume per unit time decreases. Hence the rate of the reaction decreases.

e) State TWO ways by which the rate of reaction could be increased.

f) Calculate the volume of carbon dioxide produced in the first minute at room temperature and pressure.

(Relative atomic masses: C = 12.0, O = 16.0; molar volume of gas at room temperature and pressure = 24.0 $dm^3\ mol^{-1}$)

(Edexcel GCE O Level, Paper 1, May 2009, 11(a)–(f))

18 a) The temperature used in industry needs to be high enough for the reaction to take place quickly. Explain, in terms of particles, why the rate of reaction increases when the temperature is increased.
Refer to page T34.

b) Scientists have developed catalysts which allow the reaction to take place quickly at lower temperatures. How could this be good for the manufacturer and for the environment?

(AQA GCSE (Higher Tier), Chemistry, Unit 2, Jun. 2010, 6(c)–(d))

19 A student carried out an experiment to find out the volume of carbon dioxide obtained by heating a weighed sample of magnesium carbonate.
0.418 dm^3; refer to page T34.
The student placed a 1.47 g sample of $MgCO_3$ into a test tube and heated it until there was no further change in mass.

The following reaction took place.

$$MgCO_3(s) \longrightarrow MgO(s) + CO_2(g)$$

What volume of CO_2, in dm^3, would have been given off when measured at room temperature and pressure?

(Molar mass of $MgCO_3$ = 84.3 g mol^{-1}; molar volume of gas at room temperature and pressure = 24.0 $dm^3\ mol^{-1}$)

(OCR Advanced Subsidiary GCE, Chem. A, F321, Jun. 2010, 4(c)(ii))

17 e) Any two of the following:
- Increase the temperature.
- Increase the concentration of the acid.
- Use smaller pieces of marble.

f) Mass of CO_2 produced = 0.56 g

$$\text{Number of moles of } CO_2 \text{ produced} = \frac{0.56\text{ g}}{44.0\text{ g mol}^{-1}} = 0.0127\text{ mol}$$

Volume of CO_2 produced (at room temperature and pressure)
= number of moles of CO_2
x molar volume of gas (at room temperature and pressure)
= 0.0127 mol x 24.0 $dm^3\ mol^{-1}$
= 0.305 dm^3

20 Sodium thiosulphate solution reacts with dilute hydrochloric acid. The equation for this reaction is:

$$Na_2S_2O_3(aq) + 2HCl(aq) \longrightarrow 2NaCl(aq) + H_2O(l) + SO_2(g) + S(s)$$

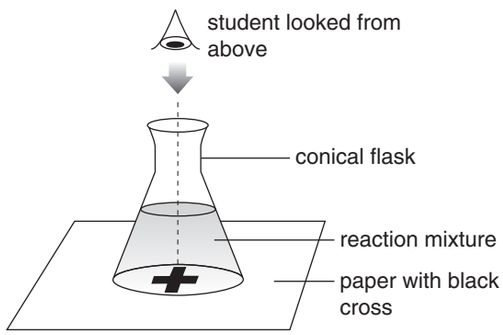
As the reaction proceeds, a pale yellow precipitate forms.

a) Identify this precipitate.

Sulphur

b) A student carried out an experiment using this method.

- She poured 25.0 cm^3 of sodium thiosulphate solution into a conical flask.
- She added 10.0 cm^3 of dilute hydrochloric acid and started a stop clock.
- She swirled the flask and placed it on a piece of paper marked with a black cross.
- She stopped the clock when she could no longer see the black cross through the reaction mixture in the conical flask.



She repeated the experiment several times using the same volume of dilute hydrochloric acid, but using different mixtures of sodium thiosulphate solution and water.

The table below shows her results.

Expt.	Volume of $Na_2S_2O_3(aq)$ (cm^3)	Volume of $H_2O(l)$ (cm^3)	Concentration of sodium thiosulphate solution (%)	Time for cross to disappear (s)
1	25.0	0	100	6.0
2	20.0	5.0	80	8.0
3	15.0		60	9.0
4	10.0	15.0	40	15.0
5	5.0	20.0	20	32.0

i) The results of *Experiment 3* are incomplete.

What volume of water should she have used for the experiment to be a fair test?

10.0 cm^3

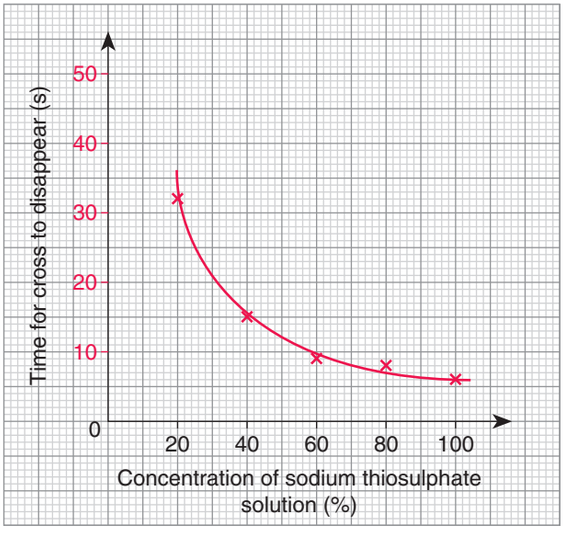
ii) The volume of dilute hydrochloric acid and the volume of the mixture of sodium thiosulphate and water were kept constant in each experiment.

State ONE other variable that she should have kept constant.

Temperature / concentration of the acid / swirling

c) Choose a suitable scale for the time taken for the cross to disappear.

Plot a graph of her results on the grid below and draw a straight line or curve of best fit.



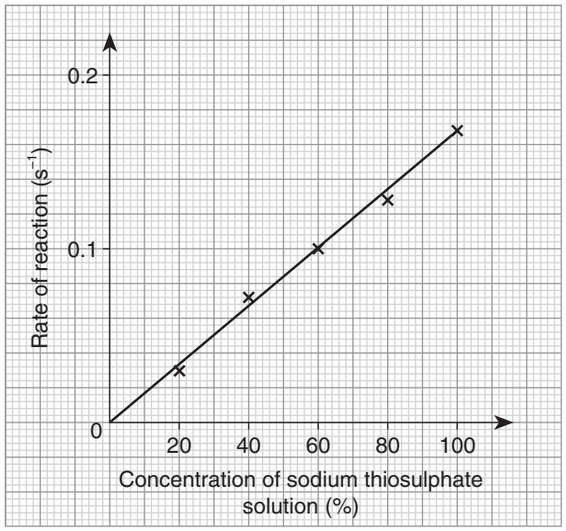
Topic Exercise

d) The student used her results to calculate the rate of the reaction at different concentrations of sodium thiosulphate solution.

The equation she used was

$$\text{rate of reaction} = \frac{1}{\text{time (in s) for cross to disappear}}$$

She then plotted a graph of rate of reaction against concentration of sodium thiosulphate solution and drew a straight line of best fit.



i) The point for 60% is incorrect.

Use the time shown in the table of results to calculate the correct rate of reaction for this concentration. Rate of reaction = $\frac{1}{9.0}$ s^{-1} = 0.11 s^{-1}

ii) The student thought the results would be least accurate at high concentrations of sodium thiosulphate solution. Suggest why this might be. The time for the cross to disappear is too short. There is a bigger error in measuring the time.

e) i) Describe the relationship shown by the graph in part (d). The rate of the reaction is directly proportional to the concentration of sodium thiosulphate solution.

ii) Explain this relationship in terms of the movement of particles.

(Edexcel IGSCE, Chemistry, Paper 3, Nov. 2010, 3)

e) ii) When the number of reactant particles per unit volume increases, there is a greater chance for collision and the number of effective collisions in a unit volume per unit time increases.

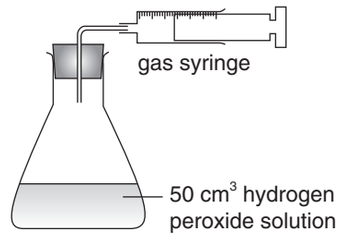
21 Hydrogen peroxide, H_2O_2, decomposes to form water and oxygen.

Refer to page T35.

a) i) Write the balanced equation for this reaction.
$2H_2O_2(aq) \longrightarrow 2H_2O(l) + O_2(g)$

ii) Describe an experiment to find the effect of temperature on the rate of decomposition of a given hydrogen peroxide solution.

Use 50 cm^3 samples of the hydrogen peroxide solution in the apparatus shown and describe what you would do and what readings you would take.



b) Alex investigated a reaction between a solid and a liquid. The reaction produced a gas.

Alex wanted to know if any of the substances X, Y or Z were catalysts for the reaction.

He carried out the reaction without any X, Y or Z. He repeated the reaction three more times under exactly the same conditions but he added a small amount of X, Y or Z.

In each case he timed how long it took for the reaction to finish.

The results are shown in the table.

Substance added	Time (s)
None	167
X	156
Y	169
Z	8

State if each of X, Y and Z acted as a catalyst and give reasons for your answer.

(Edexcel GCSE (Higher Tier), Chemistry, Unit C2, Mar. 2009, 4)

22 Hydrogen peroxide decomposes according to the following equation.

Refer to page T35.

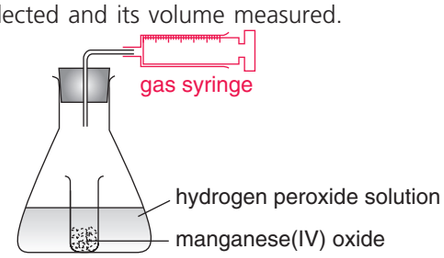
$2H_2O_2(aq) \longrightarrow 2H_2O(l) + O_2(g)$

The rate of the decomposition using manganese(IV) oxide as the catalyst was investigated by measuring the volume of oxygen formed at various times as the decomposition proceeded.

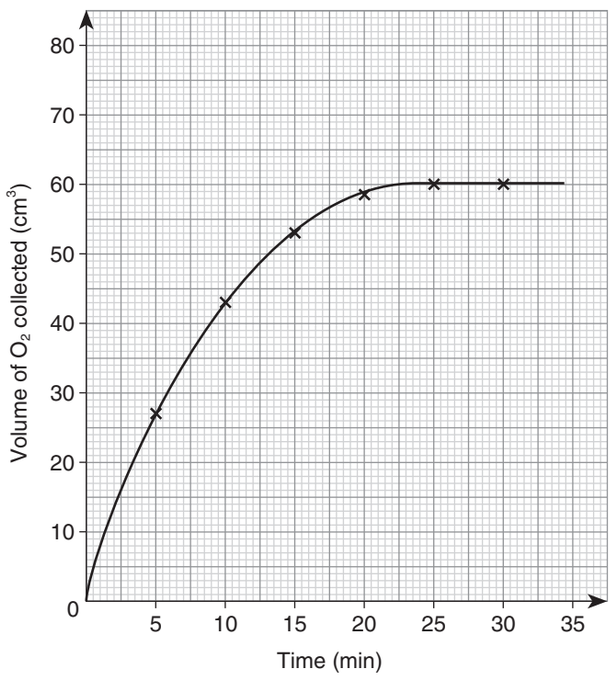
100 cm^3 of a 0.05 mol dm^{-3} hydrogen peroxide solution and 0.50 g of powdered manganese(IV) oxide were placed in the conical flask shown below.

The small glass container containing the manganese(IV) oxide was tipped over to start the decomposition and a stop watch was started at the same time.

a) Complete the diagram to show how the gas was collected and its volume measured.



b) The volume of oxygen collected at different times are shown in the graph below.



With reference to the graph above, calculate the initial rate of the reaction with respect to the volume of oxygen collected.

7.3 cm^3 min^{-1}

c) The experiment was repeated using the same amount of powdered manganese(IV) oxide, but using 150 cm^3 of 0.04 mol dm^{-3} hydrogen peroxide solution.

i) Sketch on the same graph to show the relationship between the volume of oxygen collected and time under the new set of conditions.

ii) Explain any differences between the two curves.

23 A student made the following remark:

'The rate of a gaseous reaction increases with temperature because the average kinetic energy of the reactant molecules increases with temperature.'

Is the explanation provided by the student regarding the increase in reaction rate appropriate? Elaborate your answer.

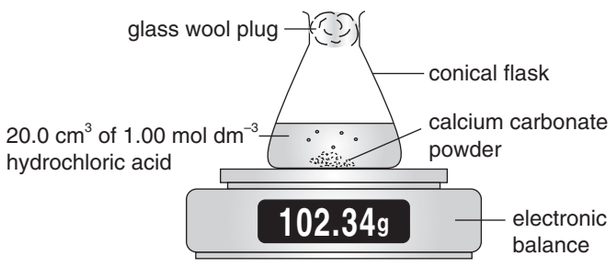
(HKASLE, Paper 1, 2010, 2)

24 Hazel investigates the reaction between hydrochloric acid and calcium carbonate.

Refer to page T36.

hydrochloric acid + calcium carbonate
$\longrightarrow$ carbon dioxide + water + calcium chloride

Look at the apparatus she uses.



Every minute she records the mass on the balance.

Look at the table of her results.

Time (min)	Mass (g)	Time (min)	Mass (g)
0	102.34	7	101.99
1	102.29	8	101.95
2	102.24	9	101.92
3	102.19	10	101.91
4	102.14	11	101.90
5	102.09	12	101.90
6	102.04	13	101.90

a) The reaction finishes after 11 minutes.

Calculate

- the total mass of carbon dioxide, CO_2, made;
  0.44 g
- the amount, in moles, of carbon dioxide made;
  0.010 mol
- the volume, in dm^3, of carbon dioxide made.
  0.24 dm^3

(Relative atomic masses: C = 12.0, O = 16.0; one mole of gas at room temperature and pressure has a volume of 24.0 dm^3.)

b) Hazel uses 20.0 cm^3 of 1.00 mol dm^{-3} hydrochloric acid.

Calculate the amount, in moles, of hydrochloric acid she uses.
0.0200 mol

c) Hazel repeats the experiment.

This time she uses 20.0 cm^3 of 1.00 mol dm^{-3} ethanoic acid and excess calcium carbonate.

i) The same volume of carbon dioxide is made. Explain why.

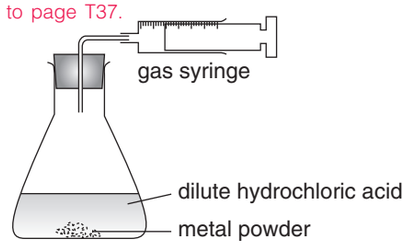
ii) The reaction is slower than with hydrochloric acid. Explain why.

Use ideas about hydrogen ions.

(OCR GCSE Gateway Science, Chem. B, Unit 2, Feb. 2012, 8)

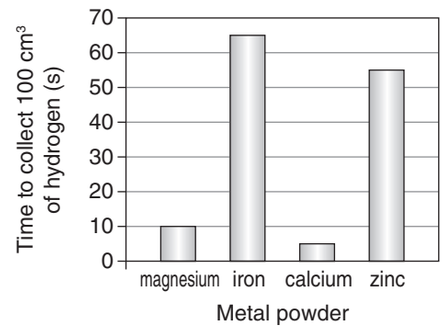
25 Cameron investigates the reaction of metal powders with dilute hydrochloric acid. Look at the apparatus he uses.

Refer to page T37.



Cameron measures the time it takes to collect 100 cm^3 of hydrogen in the gas syringe. He makes sure that all of his experiments are fair.

Look at the bar chart of Cameron's results.



a) Cameron does the experiment with zinc again.

This time he uses a more concentrated solution of hydrochloric acid. The particles in the acid are more crowded.

i) Predict how long it will take to collect 100 cm^3 of hydrogen.

ii) Explain your answer. Use ideas about collisions between particles.

b) Cameron does the experiment with zinc again. This time he uses acid at a higher temperature. The reaction is much faster. Explain why.

Use ideas about particles.

c) Cameron does the experiment with zinc again. This time he uses a lump of zinc rather than zinc powder.

What happens to the rate of reaction? Explain your answer.

d) Magnesium reacts with dilute hydrochloric acid, HCl. Hydrogen and magnesium chloride, $MgCl_2$, are made.

Write the balanced chemical equation for this reaction. $Mg(s) + 2HCl(aq) \longrightarrow MgCl_2(aq) + H_2(g)$

(OCR GCSE Gateway Science (Higher Tier), Chem. B, Unit 1, Jun. 2010, 5)

26 Safety airbags are important devices installed in vehicles. During a serious car crash, the chemicals in the airbag immediately react to release a large amount of gas. An airbag hence inflates instantly, protecting the passenger. The main chemicals in safety airbags are sodium azide (NaN_3) and potassium nitrate (KNO_3). The equations below show the reactions involved when an airbag is inflated:

$2NaN_3(s) \longrightarrow 2Na(s) + 3N_2(g)$

$10Na(s) + 2KNO_3(s) \longrightarrow K_2O(s) + 5Na_2O(s) + N_2(g)$

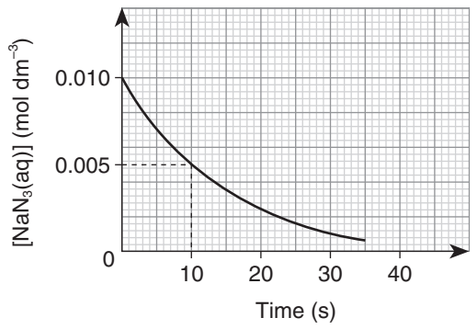
a) Explain why the $NaN_3(s)$ and $KNO_3(s)$ used in the airbags are in the form of fine powder.

b) An airbag contains 100.0 g of $NaN_3(s)$ and 200.0 g of $KNO_3(s)$. Calculate the theoretical volume, measured at room temperature and pressure, of the gas produced when the bag is inflated.

(Formula masses: $NaN_3 = 65.0$, $KNO_3 = 101.1$; molar volume of gas at room temperature and pressure = 24.0 $dm^3\ mol^{-1}$)

c) The main function of $NaN_3(s)$ is to produce $N_2(g)$ for inflating the airbags. Suggest why it is necessary to include $KNO_3(s)$ in the airbags.

d) Sodium azide is a toxic chemical. Thus any NaN_3 waste remained during the manufacture of safety airbags needs special treatment before disposal. The treatment involves first dissolving NaN_3 in water, and then reacting the solution formed with excess nitrous acid, $HNO_2(aq)$. The graph below shows the variation of the concentration of $NaN_3(aq)$ in the reaction mixture with time in one such process:



i) Calculate the average rate of consumption of $NaN_3(aq)$ in the first 10 seconds.

ii) Suggest how the instantaneous rate of consumption of $NaN_3(aq)$ at the 10th second can be determined from the graph.

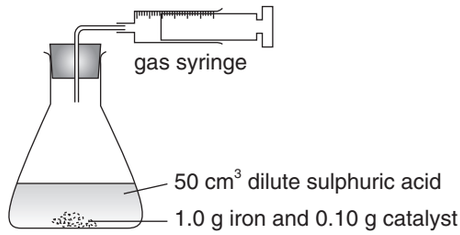
(HKDSE, Paper 1B, 2013, 11)

Topic Exercise

27 Iron reacts very slowly with dilute sulphuric acid. The reaction makes iron(II) sulphate and hydrogen.

Milly wants to make the reaction faster. She knows that catalysts make reaction faster. She tries to find a catalyst for this reaction.

Look at the diagram. It shows the apparatus she uses.



She measures the time it takes to collect 100 cm^3 of hydrogen in the gas syringe.

In *Experiments 2* to *5*, she uses 0.10 g of catalyst each time. In *Experiment 1* no catalyst is used.

Look at the results table.

Experiment	Name of catalyst	Colour of catalyst at start of reaction	Colour of catalyst at end of reaction	Mass of catalyst at the start of reaction (g)	Mass of catalyst left at the end of reaction (g)	Time to collect 100 cm^3 of hydrogen (s)
1	no catalyst added					130
2	copper powder	pink	pink	0.10	0.10	20
3	copper(II) sulphate powder	blue	pink	0.10	0.04	15
4	calcium sulphate powder	white	white	0.10	0.10	130
5	zinc powder	silver	silver	0.10	0.05	10

a) Milly thinks that copper powder is a catalyst for this reaction.

Explain how Milly made this conclusion from her results.

b) Milly uses iron powder rather than a lump of iron. This is because the reaction is faster with iron powder.

Explain why the reaction is faster. Use ideas about collisions between particles.

(OCR GCSE Gateway Science (Higher Tier), Chem. B, Unit 1, Jan. 2010, 7(b)–(c))

27 a) The experiment using copper powder as a catalyst took less time to collect 100 cm^3 of hydrogen.
The copper powder did not change colour at the end of the experiment. / The mass of copper powder remained the same at the end of the experiment.
b) The contact surface area between iron and dilute sulphuric acid increases.
There is a greater chance for collision and the number of effective collisions in a unit volume per unit time increases.